Ὅσοι ἄνθρωποι, τοσαῦται γνῶμαι ! Harmonizing Guidelines for Handwritten Text Recognition of Ancient Greek (Workshop)

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Since 2014, the Canada Research Chair on Digital Textualities[1] has led the collaborative digital edition of the *Greek Anthology* proejct[2] [Verstraete 2024, Verstraete et al. 2024, Vitali-Rosati et al. 2021, 2020; Mellet 2020]. The *codex Palatinus graecus* 23[3], a Xth century Byzantine manuscript, is the primary source for the transmission of this collection of epigrams. As part as the project, three HTR models – based on the palatinian manuscript – have been created [Guénette et al. 2024a, 2024b, 2024c]. While the creation of a model for the *Palatinus* was fairly straightforward – the manuscript is clean, clear, and contains few abbreviations – transcription choices become much more complex with less explicit manuscripts. This workshop aims to build a community of practice for transcribers of Ancient Greek manuscripts using HTR. Through this collaborative effort, a first set of guidelines will be developed to ensure consistent transcriptions across multiple document collections.

Context

In recent years, there have been significant developments in the field of HTR with the rise of several general public software such as Transkribus (Muehlberger et al. 2019) and eScriptorium (Stokes et al. 2021). In conjunction with technological progress, there has been a growing emphasis on the development of datasets and guidelines to standardize procedures and thereby facilitate their implementation. This growing interest is evidenced by a significant increase in the number of publications and conferences on the subject (Nockels et al. 2022), as well as a notable increase in the number of HTR-related initiatives, including summer schools and hackathons.[4]

For Latin script manuscripts, a critical mass of data appears to have been reached (Pinche et al. 2024, Chagué et al. 2023), paving the way for collaborative projects such as the Consortium for the Recognition of Handwritten Ancient Materials (CREMMA) (Pinche 2022) or the CATMuS project (Consistent Approaches to Transcribing Manuscripts) (Pinche et al. 2024).

However, non-Latin scripts often remain on the margins from this progress. Although promising projects have emerged, e.g. for Arabic (Kiessling 2021) and Ancient Greek (Guénette, Verstraete, Vitali-Rosati, et al. 2024, Tsochatzidis et al. 2021), these efforts remain fragmented. Currently, researchers often work on their own, following diverse practices and standards tailored to the needs of their specific projects. This situation mirrors the early days of Latin script HTR, before the development of comprehensive guidelines that slowly gained the potential to transform the field. Similarly, the establishment of a collaborative and interdisciplinary community could facilitate significant progress in non-Latin script HTR.

Thus, HTR for Ancient Greek faces several challenges, including a limited and dispersed corpus compared to Latin scripts, which hinders the development of effective models. Another issue is the lack of consistent encoding practices for Greek paleographic features. While Unicode provides basic characters for the Greek alphabet, it does not account for all the paleographic elements (such as ligatures and abbreviations) found in Ancient Greek manuscripts. This gap has led to varying encoding practices, complicating data standardization and limiting interoperability.

Current HTR approaches are often highly specialized, focusing on specific periods, regions, or ma-

nuscript types, with most projects excluding other document types like papyri. This specialization restricts the broader applicability of models. To improve data reuse and collaboration, a more generalized approach is needed, with harmonized practices across projects to enable data aggregation, sharing, and pre-annotation (Chagué et al. 2023, Hodel et al. 2021).

Aims of the Workshop

This workshop is not intended to be a technical training session on HTR, but rather a space for reflection, collaboration, and community building. It aims to bring together researchers and practitioners from a range of disciplines, including philologists, paleographers, HTR users (across various languages), and dataset creators, to address the technical and methodological challenges that have hindered the development of effective HTR for Ancient Greek.

The primary goals are:

- To foster the development of an interdisciplinary community of practice that combines complementary expertise to address the challenges posed by the automatic recognition of Greek documents.
- To develop common guidelines for the transcription and encoding of ancient Greek texts, especially for the training of automatic text recognition models.

To achieve this, the workshop will adopt a participatory, use case-driven approach. A call will be issued inviting participants to share their perspectives, experiences and challenges through lightning talks, allowing the workshop to tailor its program to their specific needs. The discussion will be rigorously documented to ensure a transparent record of the guidelines and methodologies developed during the workshop.

Results

The workshop will result in the presentation of case studies illustrating the challenges specific to the transcription of Ancient Greek manuscripts with HTR technologies. The identification of recurring issues and collaborative discussions will lead to the establishment of a preliminary set of guidelines aimed at harmonizing transcription practices for Ancient Greek manuscripts.

These initial guidelines will be made publicly available on the project's website[5], providing a shared reference point for ongoing and future work. Our longer-term goal is to establish a sustainable community of practice dedicated to refining and expanding these guidelines, ensuring they remain aligned with technological advances and the evolving needs of the field.

Notes

1. See https://www.ecrituresnumeriques.ca .
2. See https://anthologiagraeca.org .
3. The manuscript is held at the Heidelberg University Library and its digitization can be consulted at this link: .
4. See for example the TranscriboQuest 2024 < > and 2025 < > ; the Winter school of Handwritten Text Recognition of Medieval Manuscripts (Institute for Medieval Research of the Austrian Academy of Sciences, 29-30 th of January 2023) < > ; the workshop Hands-on Introduction to eScriptorium, an Open-SourcePlatform for HTR organized by Peter Stokes and Daniel Stökl Ben Ezra at DH2022 < >.
5. See .

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